Arista Access Node BAS-ACC-78 Standard Operating Procedure for BGP Route Churn, OSPF Adjacency Loss, and ISIS LSP Flooding

Node: BAS-ACC-78 | **Vendor:** Arista | **Severity:** Minor | **Alarm Code:** ALARM_3252

Description

This Standard Operating Procedure (SOP) outlines the diagnostic and resolution steps for mitigating BGP route churn, OSPF adjacency loss, and ISIS LSP flooding on Arista Access Node BAS-ACC-78 located at the Basavanagudi POP. The failure domain is primarily attributed to the inherent complexities of routing protocol interactions, specifically the dynamic nature of BGP route updates, OSPF adjacency establishment, and ISIS LSP flooding. In a multi-protocol environment, the confluence of these protocols can lead to routing instability, resulting in the observed BGP route churn, OSPF adjacency loss, and ISIS LSP flooding. This SOP aims to provide a structured approach to identifying and resolving the root causes of these issues, thereby ensuring stable routing and optimal network performance.

Pre-requisite Checks

- precondition: Ensure remote hands are dispatched to the Basavanagudi POP for physical access and assistance., justification: Physical access is required for executing diagnostic and resolution commands.

- precondition: Verify the redundant link status on BAS-ACC-78., justification: Redundant links can help mitigate the impact of routing protocol flapping.

Potential Root Causes

Cause	Probability
Misconfigured BGP neighbor settings	High
Inadequate OSPF adjacency timers	Medium
ISIS LSP flooding due to excessive routing updates	Low

Diagnostic Commands & Expected Outputs

```
Command: show bgp summary, Expected Output: BGP neighbor is UP for 00:01:30, Description: This command isolates BGP neighbor status, helping to identify potential issues with peer connectivity and routing updates.

Command: show ospf neighbor, Expected Output: Neighbor is UP for 00:01:30, Description: This command verifies OSPF adjacency status, aiding in the identification of potential issues with OSPF protocol interactions.
```

```
Command: show isis database, Expected Output: ISIS LSP flooding detected, Description: This command diagnoses ISIS LSP flooding by examining the ISIS database for excessive routing updates.
```

Resolution Steps

Step 1: Step: Reset BGP neighbor settings, Command: clear bgp neighbor , Description: This step resets BGP neighbor settings, helping to resolve misconfigured BGP neighbor settings and mitigate BGP route churn.

Step 2: Step: Adjust OSPF adjacency timers, Command: ospf timer adjacency , Description: This step adjusts OSPF adjacency timers to optimize OSPF protocol interactions and resolve OSPF adjacency loss.

Step 3: Step: Implement QoS for ISIS LSP flooding, Command: qos traffic-policy input, Description: This step implements QoS for ISIS LSP flooding, helping to mitigate the impact of excessive routing updates on the network.

Post-Check Verification

```
Command: show bgp summary, Expected Output: BGP neighbor is UP for 00:05:00, Description: This command verifies BGP neighbor status, ensuring that the resolution steps have resolved the BGP route churn issue.

Command: show ospf neighbor, Expected Output: Neighbor is UP for 00:05:00, Description: This command verifies OSPF adjacency status, ensuring that the resolution steps have resolved the OSPF adjacency loss issue.

Command: show isis database, Expected Output: No ISIS LSP flooding detected, Description: This command verifies ISIS LSP flooding, ensuring that the resolution steps have resolved the issue and mitigated the impact of excessive routing updates.
```

Roll-back Procedures

- Step: Restore original BGP neighbor settings, Command: copy running-config startup-config, Description: This step restores original BGP neighbor settings, ensuring that the network configuration is rolled back to its original state in case the fix fails.

- Step: Restore original OSPF adjacency timers, Command: copy running-config startup-config, Description: This step restores original OSPF adjacency timers, ensuring that the network configuration is rolled back to its original state in case the fix fails.

- Step: Remove QoS for ISIS LSP flooding, Command: no qos traffic-policy input, Description: This step removes QoS for ISIS LSP flooding, ensuring that the network configuration is rolled back to its original state in case the fix fails.